

Dilan Karaguler

Ph.D. Candidate in Mathematics, Michigan State University

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EDUCATION

Michigan State University

Ph.D. in Mathematics (Candidate)

– Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

East Lansing, MI, US

August 2022 – May 2027 (Expected)

Middle East Technical University (METU)

M.S. in Mathematics

– Thesis: *A Survey on Multidimensional Persistence Theory*

– Advisor: Assoc. Prof. Semra Pamuk

Ankara, Türkiye

September 2019 – September 2021

Middle East Technical University (METU)

B.S. in Mathematics, Honor Student

Ankara, Türkiye

June 2014 – June 2019

TEACHING EXPERIENCE

Lead Teaching Assistant

Department of Mathematics, Michigan State University

- Lead and mentor a team of graduate and undergraduate teaching assistants; observe in-person and online sections and give constructive feedback.
- Currently lead TA for first-year graduate students; design and run workshops that prepare them for their first teaching assignments.
- Work with faculty to set best practices for teaching assistants and prepare teaching materials for multi-section courses.

August 2023 – Present

East Lansing, MI, US

Graduate Lead, Center for Instructional Mentoring (CIM)

Department of Mathematics, Michigan State University

- Mentored first-time instructors through non-evaluative teaching feedback, one-on-one conversations and ongoing guidance during their first teaching experience.
- Created a teaching handbook built on interviews with senior TAs about the courses they taught, describing each course's expectations so junior graduate students can choose assignments that fit their needs.

Summer 2024

East Lansing, MI, US

Mentor, Academic Culture and English Skills (ACES) Program

Michigan State University

- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

Summer 2024

East Lansing, MI, US

Supervisor, Math Learning Center

Michigan State University (Teaching Exemption Fellowship)

August 2022 – May 2023

East Lansing, MI, US

Graduate Teaching Assistant

Middle East Technical University

- Led recitations and office hours, graded, and proctored online and in-person exams for Calculus I–II and Differential Equations.

January 2020 – June 2022

Ankara, Türkiye

Undergraduate Teaching Assistant

Middle East Technical University

- Graded homework and held review hours for Calculus for Mathematics Students.

March 2018 – June 2019

Ankara, Türkiye

COURSES TAUGHT

MSU MTH 299 — Transitions	Fall 2025
MSU MTH 314 — Matrix Algebra with Computational Applications	Spring 2025
MSU MTH 235 — Differential Equations	Spring 2024
MSU MTH 133 — Calculus II	Summer 2023, Fall 2023, Fall 2024
METU MTH 154 — Calculus II for Mathematics Students	Spring 2022
METU MTH 153 — Calculus I for Mathematics Students	Fall 2021
METU MTH 120 — Calculus II for Engineering Students	Spring 2021
METU MTH 119 — Calculus I for Engineering Students	Fall 2020

AWARDS, FELLOWSHIPS & SCHOLARSHIPS

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University	Summer 2026
Excellence in Teaching Award , Department of Mathematics, Michigan State University	May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University	Summer 2024
Teaching Exemption Fellowship, Michigan State University	2022 – 2023
RONESANS Scholarship, RONESANS	2018 – 2019
UNIGENC Scholarship, UNIGENC Association	2015 – 2019
Merit Scholarship, Turkish Ministry of National Education	2010 – 2014

TEACHING DEVELOPMENT & EDUCATIONAL RESEARCH

Program Evaluator & Instructional Researcher	May 2026
AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University	East Lansing, MI, US
– Designed pre/post-workshop surveys and observation protocols to evaluate program effectiveness and adult learning outcomes. – Analyzed participant feedback quantitatively and with qualitative thematic coding, tracking confidence growth across AI and semiconductor manufacturing topics. – Wrote an evaluation report grounded in adult learning theory with curriculum recommendations for the workshop.	
Colleges Online Learning Academy (COLA) Scholar	Summer 2024
Michigan State University	Portfolio
– Built a teaching portfolio website and wrote a project plan for online course design as program deliverables.	

PUBLICATIONS

D. Karaguler and G.-W. Wei. *Mayer Path Homology*. Foundations of Data Science, accepted (2026).
[doi:10.3934/fods.2026025](https://doi.org/10.3934/fods.2026025), [arXiv:2605.16525](https://arxiv.org/abs/2605.16525).

D. Karaguler. *A Survey on Multidimensional Persistence Theory*. M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians*. In preparation (2026).

TALKS

Mayer Path Homology TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU	April 2026 East Lansing, MI, US
Interaction Topology in TDA Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU	October 2025 East Lansing, MI, US
Mayer Topology in TDA Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU	October 2025 East Lansing, MI, US
From Data to Barcodes Graduate Student Seminar, Middle East Technical University	April 2021 Ankara, Türkiye

POSTER PRESENTATIONS

SIAM Great Lakes Section Meeting Illinois Institute of Technology	September 2025 Chicago, IL, US
Mathematics of Data, Dynamics, and Life Sciences Conference University of California, Irvine	March 2025 Irvine, CA, US

RESEARCH EXPERIENCE

Graduate Researcher Michigan State University	August 2022 – Present East Lansing, MI, US
– Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in <i>Foundations of Data Science</i> (with G.-W. Wei). – Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.	
Research Mentor Topology RTG, Michigan State University	May 2023 – June 2023 East Lansing, MI, US
– Mentored an undergraduate researcher on a topological data analysis project: project design, methodology and research presentation.	

INDUSTRY EXPERIENCE

Software Engineer Intern Intel Corporation	May 2026 – August 2026 Oregon, US
– Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata. – Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity. – Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices. – Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.	
Graduate Technical Intern Intel Corporation	May 2025 – August 2025 Oregon, US
– Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets. – Applied graph neural networks to search for and classify similar patterns across multi-layer designs. – Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.	

SERVICE & LEADERSHIP

Graduate Student Representative, Graduate Studies Committee

2025 – 2026

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Represented all mathematics graduate students at committee meetings with faculty.
- Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.
- Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.
- Served as a point of contact for graduate students with grievances and directed them to department and university resources.

President / Board Member

August 2022 – August 2025

Association for Women in Mathematics (AWM), MSU Chapter

East Lansing, MI, US

- Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.

Organizing Committee Member

April 2024

Graduate Student Topology and Geometry Conference, Michigan State University

East Lansing, MI, US

- Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.

President

February 2015 – December 2018

METU Math Club

Ankara, Türkiye

- Organized seminars, workshops and regular reading groups in topology, algebra and other areas.

REFERENCES

Available upon request.